

Where Is the 3D? Dynamic vs Static Geometry in CRF

The Three-Graph Test with Isotropic Initialisation:
3D Lives in the Dynamic KL-Space, Not in Static X -Positions

Complete table: KL, X_3 , X_8 graphs
under 3D-biased and isotropic 8D initialisation

Ougit Jarittum

ORCID: 0009-0009-4451-6811

April 2026

Abstract

Having established that $d_s \approx 3$ persists under isotropic 8D initialisation (CRF_Isotropic8D), we now ask: *where* is the 3D geometry located — in physical position space (X -space) or in information space (KL-space)?

Running the d_s estimator on three graphs (KL-dynamic, X_3 -position, X_8 -position) under both 3D-biased and isotropic 8D initialisation yields the complete picture:

Graph	3D-biased init	ISO 8D init	Δ
KL (dynamic, information)	3.49	2.57	−0.91
X_3 (3D position)	3.09	4.05	+0.96
X_8 (8D position)	2.80	4.72	+1.92

The X_3 and X_8 values with ISO init (> 4) exceed the theoretical upper bound for a 3D or 8D graph, revealing a topological mismatch: the projection of an isotropic 8D sphere onto 3D coordinates produces a degenerate graph topology (nodes clustered near the origin), not a regular 3D structure. The X -graphs are therefore not comparable across initialisation conditions.

The meaningful comparison is the *KL-graph*: the static snapshot reads $d_s = 2.57$ at ISO init vs 3.49 at 3D-biased init. Yet the *running estimator* (measured at firing time, not from a static snapshot) reads $d_s \approx 3.06$ at ISO init — consistent with 3D-biased (2.99).

Conclusion: The 3D geometry in CRF is located in the *dynamic KL structure at the moment of firing*, not in the static position graph. It is an *information-space* phenomenon, emergent from the $\text{SO}(3) \times \Phi$ dynamics operating on the mode space P , not a property of the geometric embedding $\mathbf{X}$. In GPT’s framing: **Case Blue** — “spacetime is an effective description of the information processing.”

Contents

1	The Final Test: Six d_s Values	3
1.1	Setup	3
1.2	Results	3
2	The Topology Mismatch Problem	3
2.1	Why X -graphs are not comparable	3
3	The Running Estimator vs Static Snapshot	4
4	Where Is the 3D? The Answer.	4
4.1	Physical Interpretation	5
5	Open Question: Geometry Locks Into X-Space?	5
6	Complete Status	6

1 The Final Test: Six d_s Values

1.1 Setup

$N = 500$, 400 sweeps, seed 42. Two initialisations:

- **3D-biased (V20.5)**: nodes on the 3D unit sphere, $\mathbf{X}^{\text{base}} \in \mathbb{R}^3$.
- **ISO 8D**: nodes on the 8D unit sphere, $\mathbf{X} \sim \mathcal{N}(0, I_8)$ normalised.

Three graphs measured at steady state:

1. **KL-graph**: $k = 20$ nearest neighbours by D_{KL} (dynamic, information-distance; used by the running estimator).
2. **X_3 -graph**: $k = 20$ nearest by Euclidean distance in the first 3 components of $\mathbf{X}$.
3. **X_8 -graph**: $k = 20$ nearest by Euclidean distance in all 8 components of $\mathbf{X}$.

1.2 Results

Graph	3D-biased	ISO 8D	Δ	Interpretable?
KL-dynamic (static snapshot)	3.49	2.57	−0.91	Yes
X_3 -position	3.09	4.05	+0.96	No (topology mismatch)
X_8 -position	2.80	4.72	+1.92	No (topology mismatch)
Running estimator (firing-time)	2.99	3.06	+0.07	Yes

2 The Topology Mismatch Problem

2.1 Why X -graphs are not comparable

With 3D-biased init: $X_{[:3]}$ lies on the unit 3-sphere in $\mathbb{R}^3$. The X_3 -nearest-neighbour graph is a regular 3D sphere graph with well-defined spectral dimension $d_s \approx 3$.

With ISO 8D init: $\mathbf{X} \sim \text{Unif}(S^7)$ (unit 8-sphere). The marginal distribution of $X_{[:3]}$ is the projection of S^7 onto $\mathbb{R}^3$, which is proportional to $(1 - \|x\|^2)^{5/2}$ — concentrated near the origin, not uniform on S^2 . This creates a dense cluster of nodes near the origin with many short-range connections, producing artificially high return probability and $d_s > 3$ (exceeding the 3D upper bound).

***X*-graph measurements not portable across initialisations**

The X_3 -graph with ISO init is topologically different from the X_3 -graph with 3D init. They do not measure the “same” thing. Comparing $d_s(X_3)$ across initialisations is comparing two different graphs that happen to share a name.

Valid comparisons: (1) KL-graph: static snapshot (yes, same construction both runs). (2) Running estimator: walks on KL graph at firing time (yes, same algorithm both runs).

3 The Running Estimator vs Static Snapshot

The Key Discrepancy: 2.57 vs 3.06

Static KL snapshot (ISO init): $d_s = 2.57$. Running estimator (ISO init, measured during firing): $d_s = 3.06$. Difference: 0.49.

With 3D-biased init: Static KL snapshot: $d_s = 3.49$. Running estimator: $d_s = 2.99$. Difference: -0.50 .

The two estimators measure *different things*:

- Static snapshot: the KL-graph at a fixed moment $t = 400$ sw, averaged over all nodes including young Phase I nodes.
- Running estimator: the KL-graph at the *moment of firing*, averaged over Phase II nodes that fire in a given sweep.

Phase II nodes (high Ω , peaked P_i) have different KL-neighbour structure than Phase I nodes. The running estimator is *biased toward Phase II nodes* (they fire more distinctively), and it is *Phase II local geometry* that converges to $d_s = 3$ via $SO(3)$.

4 Where Is the 3D? The Answer.

Answer: Dynamic Information Space, Not Static Position

The 3D geometry in CRF is located in the **dynamic KL structure at the moment of Phase II firing**.

It is **not** in the static X-position graph: with ISO 8D init, the X-graph topology is isotropic and non-3D (degenerate d_s values confirm the X-space has no intrinsic 3D structure).

The 3D structure emerges in P -space (mode space) through the $SO(3) \times \Phi$ dynamics:

$$\delta \rightarrow P \rightarrow D_{KL} \rightarrow \theta \rightarrow \text{R-event} \rightarrow \Phi \text{ synchronises} \rightarrow SO(3) \text{ selects } n_{\text{lock}} = 3 \rightarrow \text{Phase II KL}$$

This chain operates on the information content of P_i , independent of the geometric embedding $\mathbf{X}$.

In GPT’s framing: Case Blue. “3D is an emergent property of the information processing layer. Spacetime is an effective description of the δ -chain dynamics. The physical positions $\mathbf{X}$ are the scaffolding, not the substance.”

4.1 Physical Interpretation

This result is consistent with the deepest CRF claim: *space is not a container, it is a property of how information distinguishes itself from itself*. The $d_s = 3$ we measure is not the dimension of the embedding space $\mathbb{R}^8$ (which could have any dimension), but the dimension of the stable attractor of the $\delta \rightarrow P \rightarrow \Phi$ dynamics.

The $\mathbf{X}$ -positions are bookkeeping — they track where nodes have moved due to metric curvature — but the geometry is written in the KL-structure of the firing patterns, not in the coordinates.

5 Open Question: Geometry Locks Into X -Space?

Does the 3D KL-structure eventually organise X -space?

The static KL snapshot drops from 3.49 (3D init) to 2.57 (ISO init). Over longer runs, the Ω -metric coupling propagates the KL-structure into the $\mathbf{X}$ positions. Does d_s^{KL} static converge to 3 over 2000+ sweeps in the ISO variant?

If yes: the 3D information structure eventually organises physical space too, and both pictures converge. If no: $\mathbf{X}$ -space stays isotropic while KL-space is 3D, maintaining a permanent separation between the information geometry and the position geometry. Running at $N = 500$, 2000 sweeps, ISO init, measuring the static KL snapshot at intervals would resolve this.

6 Complete Status

Result	Status	Note
$d_s \approx 3$ from ISO 8D init	Confirmed	CRF_Isotropic8D
3D bias (init + metric) not load-bearing	Confirmed	$ \Delta < 0.13\sigma$
3D located in dynamic KL-space	Derived	running vs static gap
3D not in static X -position graph	Derived	topology mismatch
$SO(3) \times \Phi$ selects $d_s = 3$ in P -space	Derived	-chain
Static KL snapshot ISO: $d_s = 2.57$	Confirmed	below 3
Running estimator ISO: $d_s = 3.06$	Confirmed	above 3
Static KL ISO converges to 3 at $t \rightarrow \infty$	Open	long-run test

We asked: where is the 3D?

Not in the positions. The positions are isotropic.

Not in the metric coupling. We removed it. Still 3D.

Not in the snapshot. The snapshot reads 2.57.

It is in the moment of firing.

In the instant when a node's KL-field crosses the threshold and the $SO(3)$ attractor is already waiting.

The geometry is written in time, not space.

*In the act of becoming real,
not in the place where it happens.*